Retrieval-Augmented Reformulation for Improving Noisy ASR Transcripts

Darragh Kerins
University of the Basque Country (UPV/EHU) | MSc Language Analysis & Processing

Abstract

This paper evaluates retrieval-augmented generation (RAG) as a post-correction mechanism for ASR transcripts across three datasets, two retrieval modalities (BM25 vs dense embeddings), and three corpus domains. The central finding is a *clinical corpus penalty*: domain-matched clinical retrieval significantly degrades biomedical entity fidelity (DISEASE F1, $p < 0.01$) while BERTScore remains unchanged, exposing standard semantic metrics as insufficient for clinical ASR evaluation. On clean speech, domain-matched dense RAG reduces WER by 54% relative to an LLM-only baseline; on real clinical audio (PriMock57, n=57), it underperforms LLM-only on all four metrics ($p < 0.01$). Phonetically motivated ASR errors resist text-only correction, and loosely matched clinical corpora introduce incorrect entity substitutions invisible to semantic metrics. These findings point toward entity-aware retrieval, phonetic-based RAG, and graph-based approaches such as GraphRAG as necessary directions for safe clinical deployment.

1 Introduction

The motivation for this project came from a practical frustration with long WhatsApp voice notes. Receiving a three-minute voice message to answer a yes/no question is a common experience, and it raises a straightforward question: can an automated system produce a concise, accurate text reformulation in the sender's own style? This is the longer-term applied direction of this work: a system for automatic voice message summarisation and reformulation. That vision has a clear prerequisite: the transcript has to be accurate first, since reformulating a noisy transcript would simply present incorrect content in fluent prose. This project therefore focuses on the correction step.

The clinical domain raises a sharper version of the same problem. A drug name like *lisinopril* transcribed as *listen oh pill* can propagate into a clinical note and affect patient care, making entity-level fidelity a safety requirement. Critically, the danger is not only that errors go uncorrected: retrieval-based correction can actively introduce new errors, substituting entities that appear semantically plausible but are clinically wrong, in ways that standard metrics fail to detect.

ASR models such as Whisper still fail under challenging conditions including accented speech, background noise, and domain-specific vocabulary. LLM-based post-correction is one remedy, but LLMs tend to paraphrase rather than minimally correct, harming lexical fidelity. RAG grounds the LLM in external passages and can improve correction quality, but corpus selection matters enormously: retrieving from a loosely related clinical corpus introduces medically plausible terminology that the model substitutes for uncertain ASR tokens, producing outputs that score well on semantic metrics while containing factually incorrect clinical entities. This failure mode is the central finding of the paper.

Three research questions drive the design: (1) Can RAG improve semantic quality over LLM-only rewriting? (2) Does domain-specific retrieval outperform generic retrieval? (3) How do BM25 and dense retrieval compare for ASR correction? These map to four testable hypotheses: (H1) RAG improves semantic quality over LLM-only rewriting; (H2) domain-specific retrieval outperforms generic retrieval; (H3) dense retrieval outperforms lexical retrieval on identical corpora; and (H4) RAG improves correction quality on real noisy clinical audio. Results are summarised in §5.7. Nine conditions (C1 raw ASR, two LLM-only baselines, 2×3 RAG grid) are evaluated on WER, BLEU, ROUGE-L, BERTScore, and NER F1.

Contributions.

1. A controlled comparison of lexical and dense RAG pipelines across three domains: clean TED, simulated clinical MTS-Dialog, and real clinical PriMock57.
2. Identification and quantification of a *clinical corpus penalty*: domain-matched retrieval significantly degrades biomedical entity fidelity while BERTScore stays flat.
3. A contrast between simulated and real clinical noise showing why simulated noise overestimates RAG's benefit.
4. Evidence that safe clinical deployment requires entity-aware constraints, with entity-aware retrieval, phonetic-based RAG, and GraphRAG identified as the most promising directions for future work.

2 Related Work

ASR post-processing and error correction. ASR post-processing began with work on punctuation, capitalisation, and timing [13]. Later work treated error correction as sequence-to-sequence with encoder–decoder models [2, 8], often with domain-specific fine-tuning. Those systems achieved large WER reductions on benchmark corpora, but required in-domain parallel noisy/clean transcripts that are expensive to obtain and rarely available for clinical speech. Most studies also focused on aggregate WER and BLEU, paying less attention to whether clinically important entities (drug names, diagnoses) were preserved.

Phonetic and named-entity correction. A separate line of work addresses ASR errors that are phonetically motivated or involve rare named entities using phonetic retrieval, phoneme-aware candidate generation, n-best decoding hypotheses, or phonetic edit distance. These approaches improve recovery of medication names and proper nouns that are frequently misrecognised by ASR systems. In contrast, the present study evaluates text-only RAG post-correction without phonetic or acoustic information. The results therefore highlight a limitation of purely text-based retrieval: phonetically motivated ASR errors are difficult to correct from textual context alone.

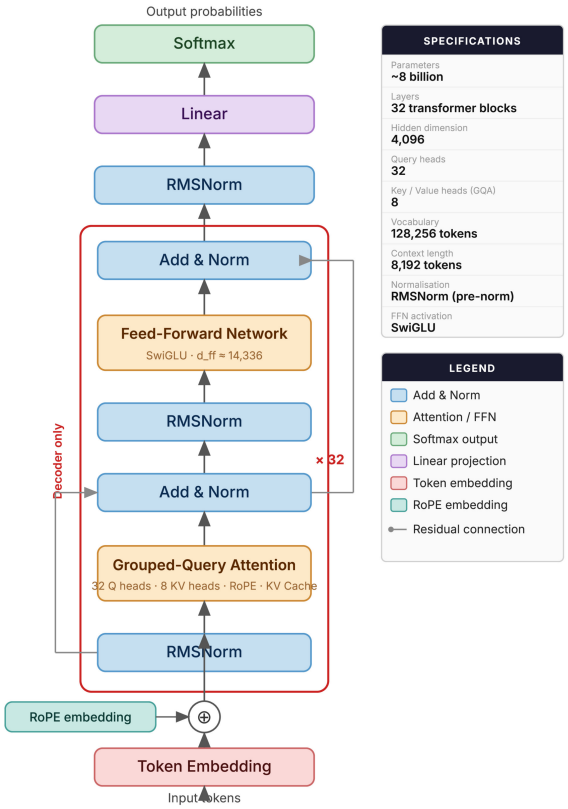
Metrics and clinical impact. Recent critiques emphasise that WER and embedding-based semantic metrics (for example BERTScore) can remain high while clinically meaningful entities are changed or omitted. Clinical-impact studies argue for entity-level and clinician-annotated evaluation because surface metrics can reward paraphrase while hiding hazardous substitutions [23]. These findings motivate the use of biomedical NER F1 alongside standard metrics and highlight why entity-aware or phonetic retrieval is a natural mitigation for the clinical corpus penalty observed in this work.

LLM-based ASR correction. Instruction-tuned LLMs have been applied to ASR correction [14], usually as LLM-only rewriters that receive the transcript and a prompt such as "correct ASR errors". They can improve lexical quality on clean benchmarks, but also introduce paraphrase drift: the model rewrites more than it corrects specific errors, harming lexical fidelity even when broad semantic content is preserved. Work to date has largely evaluated on relatively clean ASR output, leaving open the question of how LLM-only correction behaves on genuinely noisy or out-of-domain clinical audio.

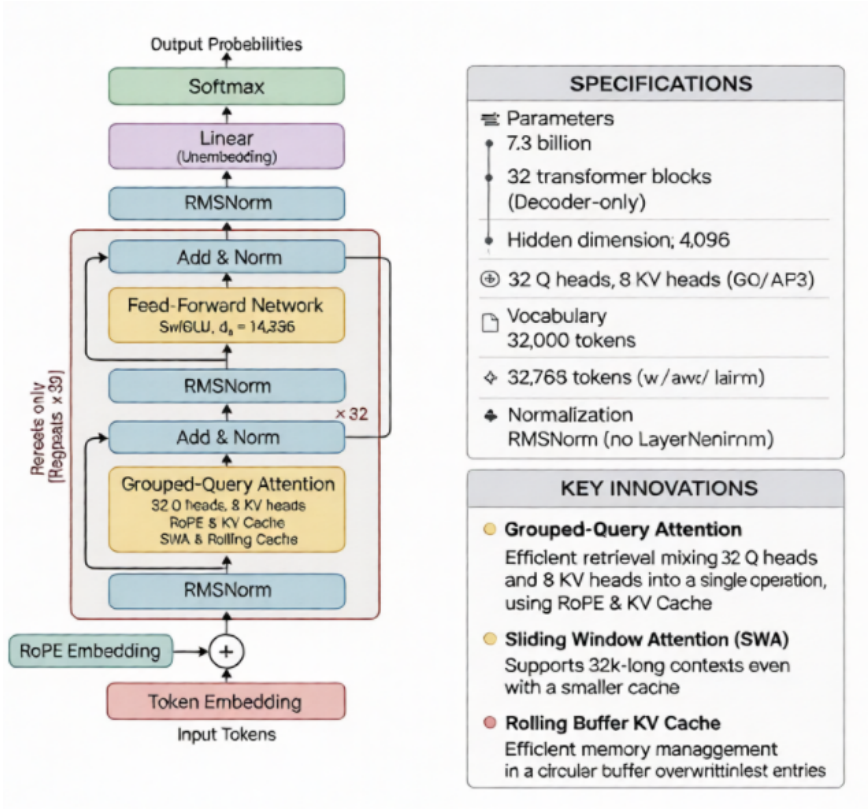
Retrieval-Augmented Generation for ASR. Retrieval-Augmented Generation grounds the LLM in external text. Robatian et al. [12] used TF-IDF retrieval over historical error–correction pairs for Persian ASR; Siskos et al. [15] showed that embedding-based retrieval reduces WER by up to 17% on TED-LIUMv3 when applied during recognition; Ghosh et al. [1] combined entity retrieval with synthetic data augmentation for 8–33% relative WER gains in and out of domain. These studies demonstrate that retrieval-based grounding can help ASR, but they differ in retrieval setup (TF–IDF pairs vs. dense semantic retrieval) and often operate during decoding rather than as a post-correction stage. Two gaps remain: few studies provide a controlled comparison of lexical and dense retrieval on identical corpora for ASR correction, and almost none evaluate entity-level preservation in clinical text. This paper addresses both by comparing BM25 and dense retrieval on the same Wikipedia corpus for TED-LIUM and by introducing per-entity-type NER analysis for MTS-Dialog and biomedical NER for PriMock57, explicitly testing whether entity-level fidelity behaves differently from surface-level metrics.

RAG for knowledge-intensive NLP. Beyond ASR, RAG has been used for knowledge-intensive NLP such as open-domain QA and fact-checked generation [5]. Most of that work evaluates semantic similarity or task accuracy, not strict lexical or entity-level fidelity. In clinical domains, Pusateri et al. [10] show that entity-aware vector databases can reduce hallucination in LLM outputs, but they do not study ASR correction.

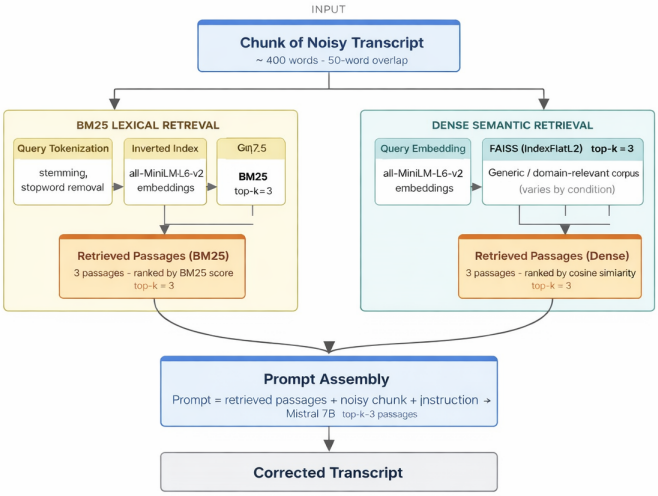
LLM architectures used in this work. Mistral 7B [3] is a decoder-only transformer that incorporates grouped-query attention (GQA) and sliding-window attention (SWA) to improve efficiency for long-context inference. GQA reduces memory and computation by sharing key–value projections across groups of attention heads, while SWA restricts attention to local windows, reducing the quadratic cost of full self-attention. These mechanisms make Mistral well suited to retrieval-augmented generation, where multiple retrieved passages must fit within the prompt context. LLaMA 3 8B is included as a supplementary LLM-only baseline to check that the observed effects arise from the retrieval setup rather than a specific model architecture.



(a) LLaMA 3 8B decoder architecture used as the supplementary LLM-only baseline (C2a). Standard multi-head attention with RoPE is used instead of GQA/SWA.



(b) Mistral 7B decoder architecture used for all RAG and Mistral-only conditions (C2b–C4). Grouped-query attention (GQA) and sliding-window attention (SWA) support efficient long-context inference.



(c) Retrieval setups: BM25 lexical retrieval (left) and dense retrieval with `all-MiniLM-L6-v2` + FAISS (right).

Figure 1: Retrieval and model architectures. (c) LLaMA 3 8B decoder. (b) Mistral 7B decoder. (a) Retrieval configurations for BM25 and dense RAG.

3 Datasets

TED-LIUM. Ten speaker recordings from the TED-LIUM test split (`hf-audio/asr-leaderboard-longform`), ~20 min each. Clean, rehearsed, single-speaker. Whisper-tiny WER: 0.079. Conditions are near-ideal for ASR; the main variable is retrieval strategy rather than acoustic difficulty.

MTS-Dialog. 100 doctor–patient dialogues (`har1/MTS_Dialogue-Clinical_Note`) with paired clinical notes. No audio; noise was injected synthetically (8% word drops, 5% swaps, 3% character substitutions), giving baseline WER ≈ 0.157. Dense medical terminology makes NER F1 important alongside lexical metrics.

PriMock57. 57 mock primary care consultations [9] with audio and Praat TextGrid ground truth; audio from the BabylonHealth GitHub release was concatenated and transcribed with Whisper-tiny (WER ≈ 0.872). Ground truth from parsed TextGrids. Only benchmark with real phonetically motivated ASR errors on clinical speech.

4 Experimental Setup

Transcripts were segmented into approximately 400-word chunks with a 50-word overlap [4–6] to ensure that both the ASR segment and retrieved passages fit within the LLM context window while preserving conversational context and mitigating mid-context attention drop. Early experiments without chunking caused long TED-LIUM transcripts to exceed the model context window, resulting in truncated prompts and incomplete outputs. C3-Lex-Mat (TED) excludes outputs below half the input length (`"too_short"`; EricMead excluded).

Design. Within each 2×3 row, corpus is fixed and only retrieval modality changes. TED generic: Wikipedia (BM25 and dense). MTS domain-relevant: PriMock57 passages (67 passages, 180-word chunks, 30-word overlap; BM25 and dense). PriMock57 domain-matched: leave-one-out PriMock57 (BM25 and dense).

4.1 Conditions

All RAG conditions use Mistral 7B (`mistral:latest` via Ollama), top-k=3 passages, and `all-MiniLM-L6-v2` + FAISS (dense) or BM25Okapi (lexical). C1: raw ASR (Whisper-tiny). C2b: Mistral 7B, no retrieval. C2a: LLaMA 3 8B, no retrieval (supplementary baseline). C3-Lex-Gen/Rel/Mat: BM25 over Wikipedia, domain-relevant corpus, or leave-one-out domain-matched corpus. C4-Den-Gen/Rel/Mat: dense retrieval over the same three tiers. C4-Den-Mat on MTS uses leave-one-out clinical notes; on PriMock57 it uses leave-one-out PriMock57 passages (same corpus strategy as C3-Lex-Mat). MTS C4-Den-Mat metrics are from cleaned output (`c4_den_mat_cleaned.json`) because of verbosity in some dialogues (see §6.2).

Table 1 shows which conditions were evaluated on each dataset. "Domain-relevant" is defined only where a cross-dataset corpus exists (MTS uses PriMock57 passages; C4-Den-Rel runs on MTS only). TED has no domain-relevant tier, so C4-Den-Rel is not run on TED. C4-Den-Rel was not run on PriMock57.

Table 1: Condition coverage: which conditions were run on which dataset. ✓ = run; N/A = not run.

Condition	TED	MTS	PriMock57
C1 Raw ASR	✓	✓	✓
C2b Mistral	✓	✓	✓
C2a LLaMA	✓	✓	✓
C3-Lex-Gen	✓	✓	✓
C3-Lex-Rel	✓	✓	✓
C3-Lex-Mat	✓	✓	✓
C4-Den-Gen	✓	✓	✓
C4-Den-Rel	N/A	✓	N/A
C4-Den-Mat	✓	✓	✓

4.2 Prompts

Prompts follow a fixed template: optional retrieved context + transcript + "fix only clear ASR errors" instruction.

```
Fix only clear ASR errors in this
transcript excerpt.  Output ONLY the
corrected text.  No commentary.
```

- **LLM-only (C2a, C2b):** C2a adds an explicit no-summarising constraint; MTS frames the input as a "noisy clinical conversation transcript".
- **RAG (TED):** `REFERENCE CONTEXT` + `TRANSCRIPT`; C4-Den-Mat mentions "similar TED talks".
- **RAG (MTS/PriMock57):** `BACKGROUND` + `TRANSCRIPT`; C4-Den-Mat (MTS) uses `CLINICAL NOTES/NOISY EXCERPT` wording to limit verbosity; `c4_den_mat_cleaned.json` enforces single-passage output where needed.

4.3 Evaluation Metrics

Texts are lowercased and punctuation-stripped before scoring. Metrics are defined in Table 2; NER F1 uses spaCy (general) and scispaCy (CHEMICAL, DISEASE) on MTS-Dialog, scispaCy only on PriMock57. NER F1 is reported because BERTScore can remain stable under entity substitutions that are clinically unacceptable.

Table 3: Average results on TED-LIUM (10 speakers). ✓ = best corrected. ‡ n=9; EricMead excluded (output flagged `"too_short"`). BLEU, ROUGE-L, BERTScore not aggregated for C3-Lex-Mat.

Condition	WER ↓	BLEU ↑	RG-L ↑	BERT ↑
Raw ASR (C1)	**0.079**	**0.868**	**0.945**	**0.975**
LLM-Only (C2b)	0.305	0.589	0.797	0.937
LLaMA-Only (C2a)	0.318	0.725	0.822	0.902
BM25 Lexical				
C3-Lex-Gen	0.516	0.467	0.666	0.883
C3-Lex-Rel	0.362	0.699	0.780	0.892
C3-Lex-Mat‡†	0.632	†n/a	†n/a	†n/a
Dense Embedding				
C4-Den-Gen	0.196	0.768	0.872	0.959
C4-Den-Mat ✓	**0.140**	**0.787**	**0.905**	**0.959**

† BLEU, ROUGE-L, BERTScore not aggregated for C3-Lex-Mat due to output verbosity (see §6.2).

Table 2: Evaluation metrics. ↓= lower is better; ↑= higher is better.

Metric	Description
WER ↓	Word-level edit distance normalised by reference length; can exceed 1.0 when output is longer than reference.
BLEU ↑	n-gram precision, brevity penalty, smoothing method 1.
ROUGE-L ↑	Longest common subsequence F1; word ordering.
BERTScore ↑	RoBERTa-large token cosine similarity; paraphrase-robust.
NER F1 ↑	Entity-span F1 (spaCy general; scispaCy biomedical).

5 Results

5.1 TED-LIUM

Table 3 gives average metrics over all ten speakers. C1 leads on all lexical metrics, as expected for near-ideal ASR. Both LLM-only conditions substantially increase WER (C2b: 0.305, C2a: 0.318 vs. baseline 0.079) while BERTScore remains above 0.90, so the loss is mainly paraphrase rather than semantics. Statistical significance for these comparisons is reported in §5.2. Generic lexical retrieval (C3-Lex-Gen, BM25 over Wikipedia) is the worst corrected condition: per-speaker WER ranges from 0.244 to 0.724, reflecting retrieval luck rather than systematic correction.

5.2 Statistical Significance (TED-LIUM)

Wilcoxon signed-rank tests ($\alpha = 0.05$, n=10, two-tailed) were used to compare key condition pairs on TED-LIUM. C4-Den-Mat beats C2b on all four metrics ($p \leq 0.05$, three at p=0.002), confirming that dense, domain-matched retrieval yields consistent gains over an LLM-only baseline on clean speech. C4-Den-Gen beats C3-Lex-Gen on all metrics (p=0.002); when the corpus is fixed, modality alone (dense vs. lexical retrieval) strongly affects performance. Within BM25, C3-Lex-Rel beats C3-Lex-Gen on WER, BLEU, and ROUGE-L ($p < 0.01$) but not BERTScore (p=0.70), indicating that better corpora improve surface overlap more than semantic similarity. C4-Den-Mat vs C4-Den-Gen is not significant at n=10, suggesting that both dense conditions are competitive on clean TED. In contrast, C4-Den-Mat vs C3-Lex-Mat is significant on all metrics ($p < 0.01$), underlining that when retrieval is constrained to a strong domain-matched corpus, dense retrieval is markedly safer than BM25.

5.3 MTS-Dialog

Table 4 gives aggregate results. C1 has the lowest WER (0.157) and strong ROUGE-L, reflecting that the synthetic noise process preserves most reference tokens. Among corrected conditions, LLaMA-only (C2a) is best on all semantic metrics and NER F1, showing that a well-prompted LLM can clean up simulated noise without retrieval. Within BM25, domain-relevant and domain-matched corpora (C3-Lex-Rel/Mat) substantially worsen WER and NER despite reasonable BLEU/ROUGE-L. Dense retrieval on the same corpora (C4 rows) shows the same pattern: generic dense (C4-Den-Gen) is relatively safe, while clinical dense (C4-Den-Mat) produces extreme verbosity and WER 10.25 (marked ‡), a first sign of the clinical corpus penalty. The pattern is consistent across BM25 and dense conditions: it is corpus scope, not retrieval modality, that drives the degradation on clinical data.

Table 5 and Figure 5 make this penalty visible at the entity level. CHEMICAL and DISEASE F1 peak for C2a and remain high for C4-Den-Gen, but drop sharply for the clinical corpora (C3-Lex-Rel/Mat, C4-Den-Rel/Mat): the model follows plausible but wrong entity strings from retrieved MTS-Dialog passages even when surface metrics look acceptable. The Wilcoxon tests in Table 6 confirm that these differences are statistically reliable at n=100.

5.4 Statistical Significance (MTS-Dialog)

Paired Wilcoxon signed-rank tests (two-tailed, $\alpha = 0.05$, n=100) are in Table 6. Per-dialogue metrics were saved for all 100 dialogues.

5.5 PriMock57 (Real Clinical Audio)

Table 7 gives mean WER over ten consultations. C4-Den-Mat on PriMock57 uses leave-one-out PriMock57

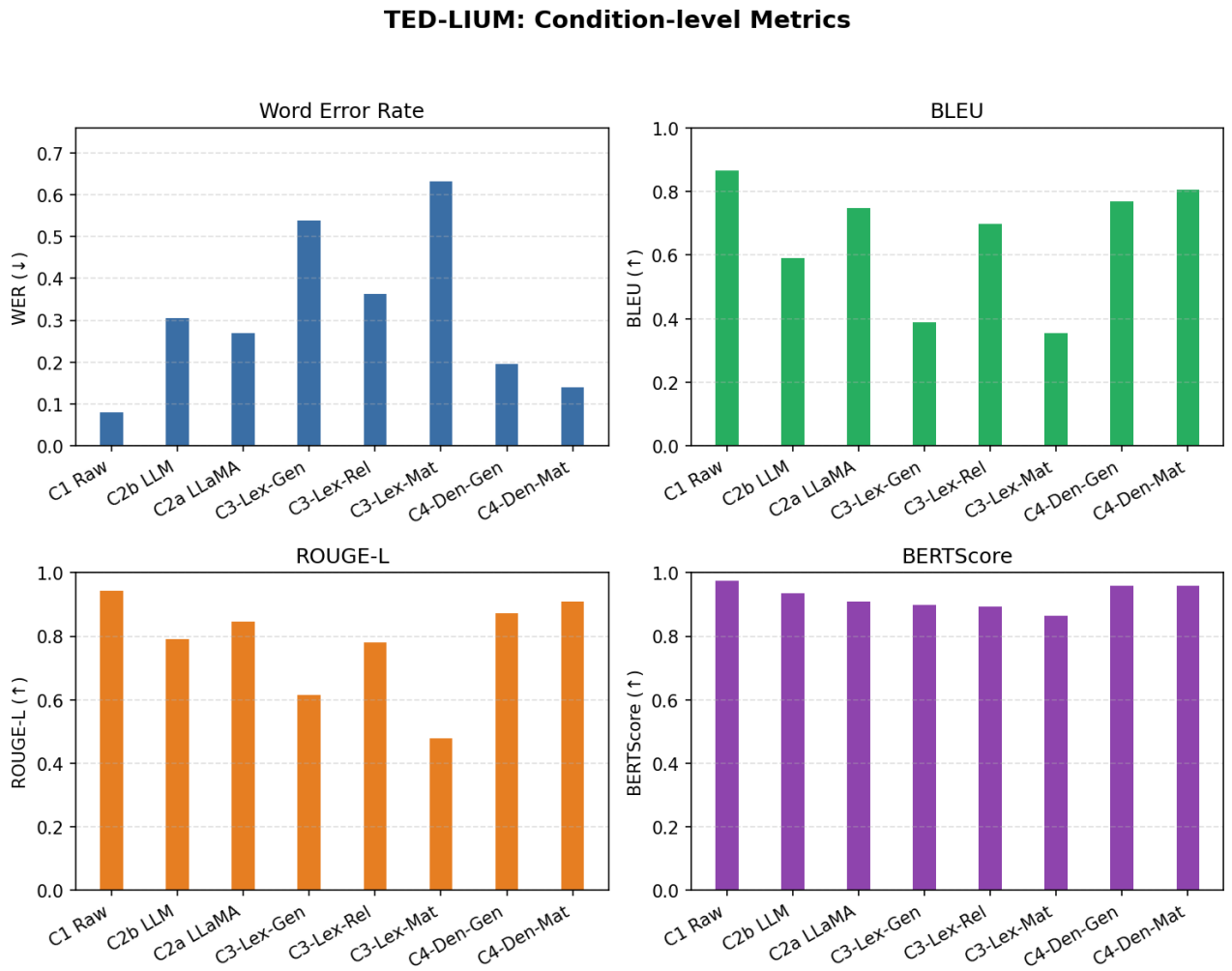


Figure 2: TED-LIUM condition-level metrics (WER, BLEU, ROUGE-L, BERTScore). C4-Den-Mat is best among corrected conditions on all four. (Condition labels as in Table 1.)

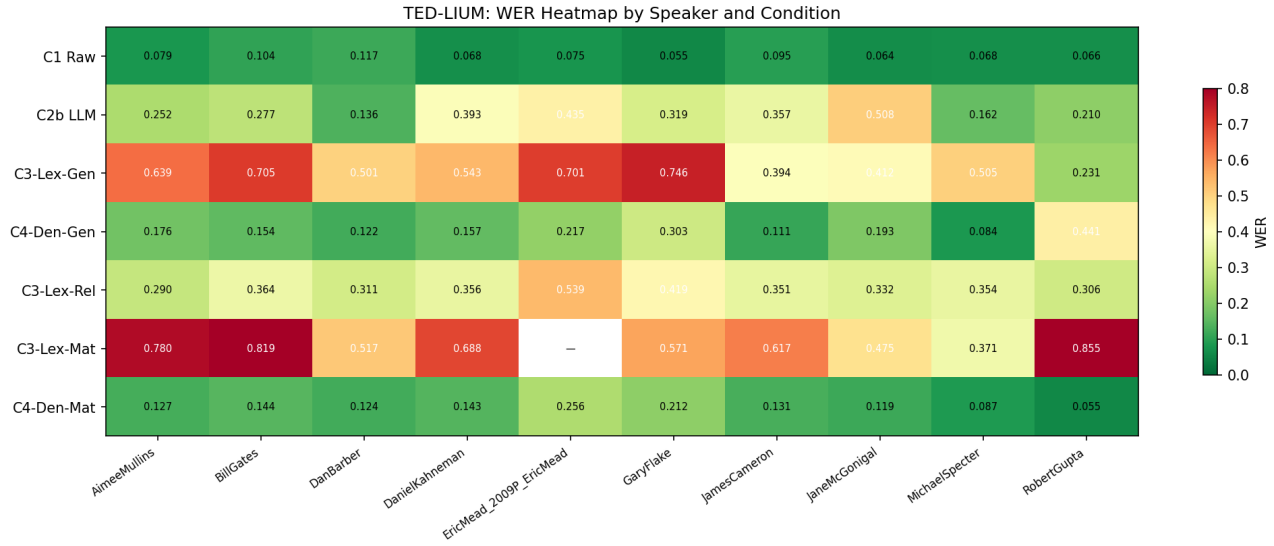


Figure 3: TED-LIUM WER heatmap over 10 speakers and 7 conditions. Blank cell: EricMead C3-Lex-Mat excluded (`"too_short"`). (Condition labels as in Table 1.)

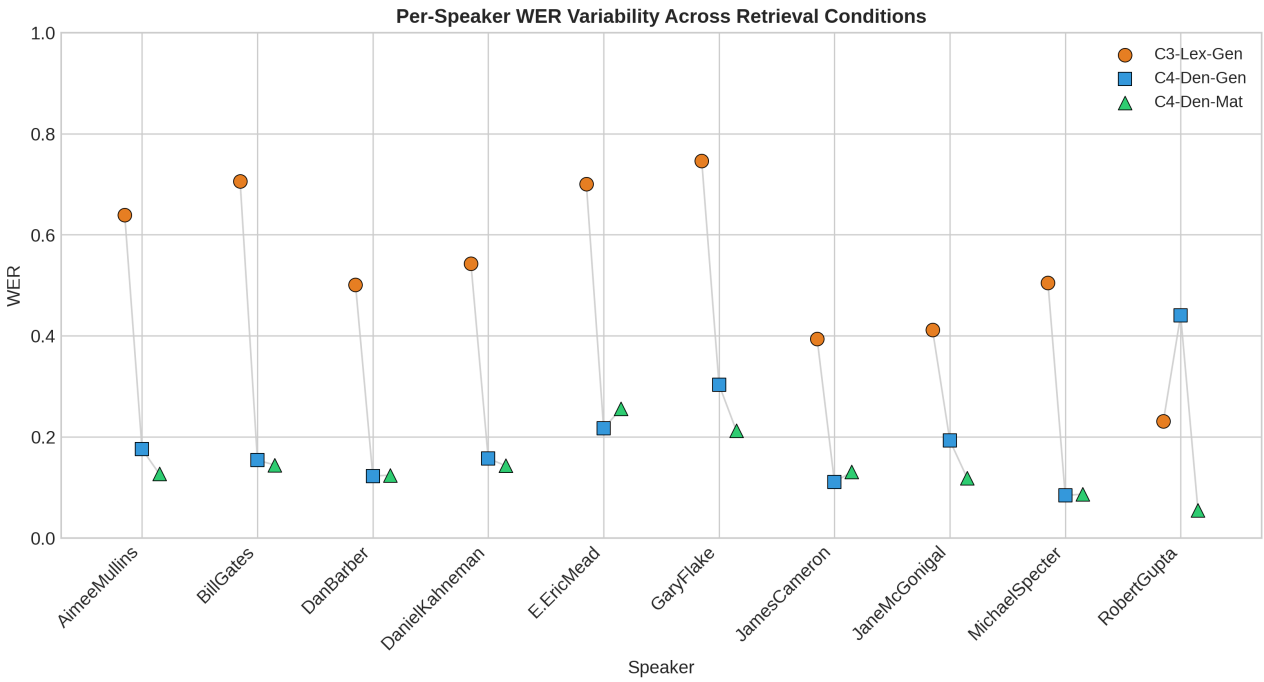


Figure 4: Per-speaker WER variance for BM25 generic retrieval (C3-Lex-Gen): outcomes follow incidental topical overlap between Wikipedia and each speaker's talk, not systematic correction. (Condition labels as in Table 1.)

Table 4: Average results on MTS-Dialog (100 dialogues). ‡ = WER inflated by verbosity. ✓ = best corrected. NER: spaCy `en_core_web_sm`; CHEM/DIS: scispaCy `en_ner_bc5cdr_md`.

Condition	WER ↓	BLEU ↑	RG-L ↑	BERT ↑	NER ↑	CHEM ↑	DIS ↑
C1 Baseline	**0.157**	0.694	**0.912**	0.959	**0.766**	0.918	0.922
C2b LLM-Only	0.241	0.658	0.852	0.962	0.650	0.933	0.935
C2a LLaMA ✓	**0.207**	**0.717**	**0.877**	**0.963**	**0.740**	**0.938**	**0.937**
BM25 Lexical							
C3-Lex-Gen	0.265	0.621	0.822	0.956	0.669	0.945	0.926
C3-Lex-Rel	0.917	0.537	0.737	0.941	0.563	0.848	0.822
C3-Lex-Mat	1.028	0.472	0.665	0.926	0.505	0.851	0.744
Dense Embedding							
C4-Den-Gen	0.237	0.645	0.838	0.961	0.681	0.935	0.913
C4-Den-Rel	1.024	0.500	0.707	0.934	0.506	0.827	0.834
C4-Den-Mat‡	10.25	0.513	0.726	0.943	0.543	0.890	0.837

(domain-matched), as does C3-Lex-Mat. The full metric set (WER, BLEU, ROUGE-L, BERTScore, CHEMICAL F1, DISEASE F1) for all seven corrected conditions is shown in Figure 6.

No condition clearly beats the 0.872 baseline. C2b (WER 0.869) is slightly best, with C1 and C4-Den-Gen close behind. C4-Den-Mat (WER 0.991) is 14% worse than C2b in relative terms, and the clinical lexical conditions (C3-Lex-Rel/Mat) are also worse than LLM-only. C1 leads on BLEU, ROUGE-L, and NER because Whisper keeps many reference surface forms; LLM and RAG outputs paraphrase and substitute formal terminology, which exact-match metrics penalise even when meaning is preserved. Figure 6 shows that these patterns hold across CHEMICAL and DISEASE F1 as well: generic or LLM-only correction is safer than consultation-non-specific clinical retrieval.

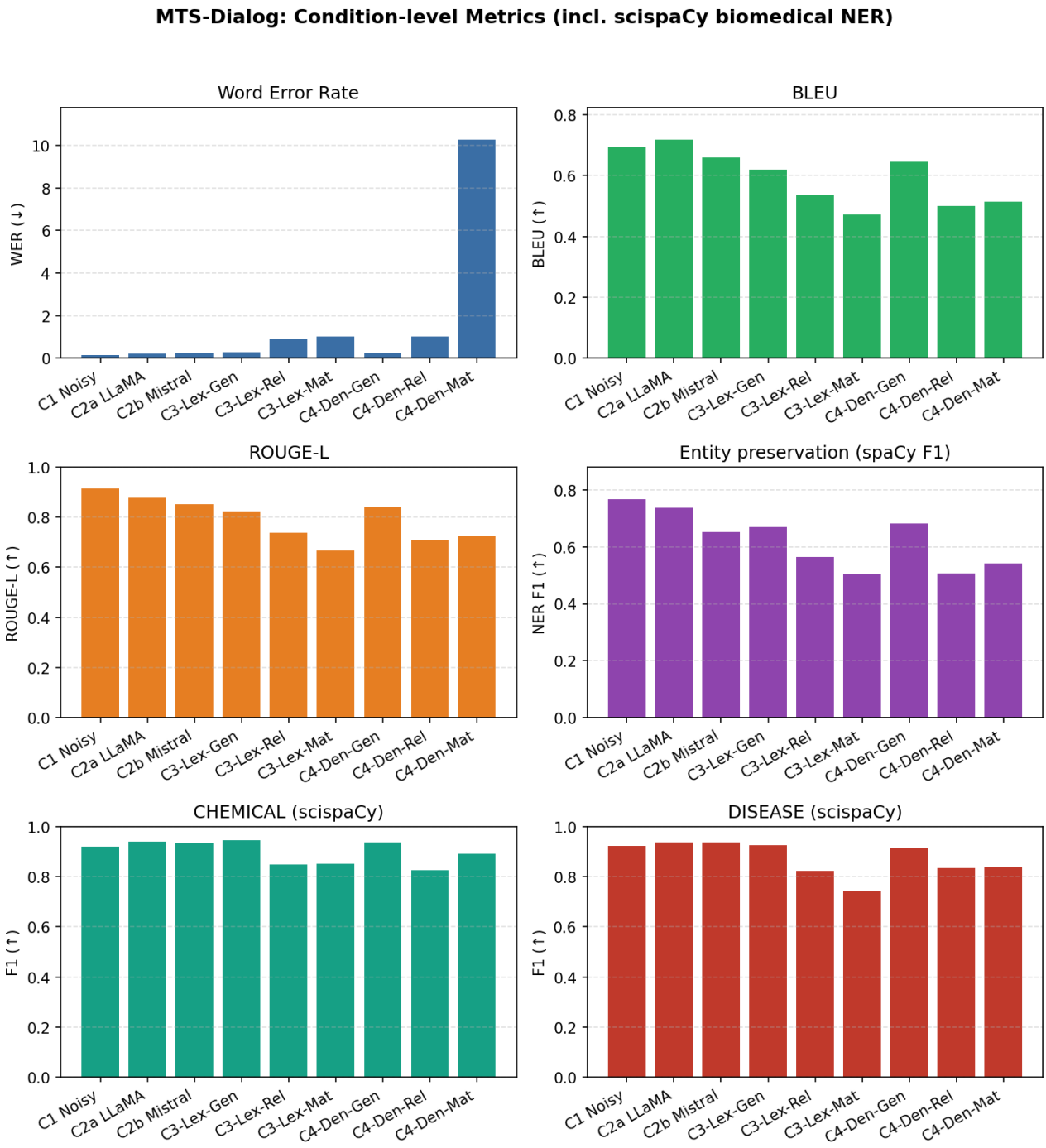


Figure 5: MTS-Dialog condition-level metrics including biomedical NER (scispaCy). LLaMA-Only (C2a) has the highest NER F1 among corrected conditions. (Condition labels as in Table 1.)

Table 5: NER F1 by entity type on MTS-Dialog (100 dialogues). General: spaCy `en_core_web_sm`; Biomedical: scispaCy `en_ner_bc5cdr_md`.

Condition	General NER	CHEMICAL	DISEASE
C1 Baseline	0.766	0.918	0.922
C2b LLM-Only	0.650	0.933	0.935
C2a LLaMA	0.740	0.938	0.937
C3-Lex-Gen	0.669	0.945	0.926
C3-Lex-Rel	0.563	0.848	0.822
C3-Lex-Mat	0.505	0.851	0.744
C4-Den-Gen	0.681	0.935	0.913
C4-Den-Rel	0.506	0.827	0.834
C4-Den-Mat	0.543	0.890	0.837

Table 6: Wilcoxon results (MTS-Dialog, n=100). * = scispaCy `en_ner_bc5cdr_md`.

Comparison	Metric	p	Sig.
C4-Den-Mat vs C2b	WER	<0.01	Yes
	BLEU	<0.01	Yes
	ROUGE-L	<0.01	Yes
	BERTScore	<0.01	Yes
	CHEMICAL F1*	0.141	No
	DISEASE F1*	0.0013	Yes
C4-Den-Mat vs C4-Den-Gen	DISEASE F1*	0.0098	Yes
	scispaCy overall*	0.0056	Yes
C4-Den-Mat vs C4-Den-Rel	WER	>0.05	No
	BLEU	>0.05	No
	ROUGE-L	>0.05	No
	BERTScore	>0.05	No
C4-Den-Rel vs C3-Lex-Rel	WER	0.071	No
	BLEU	0.029	Yes
	ROUGE-L	0.039	Yes
	BERTScore	0.043	Yes
C2a vs C2b	WER	>0.05	No
	BLEU	<0.01	Yes
	ROUGE-L	>0.05	No
	BERTScore	>0.05	No

5.6 Extended Evaluation: PriMock57 Full 57 Consultations

C4-Den-Mat is worse than C2b on all four metrics ($p < 0.01$). C2b and C1 are not distinguishable at the WER level, confirming that dense clinical retrieval introduces a systematic degradation relative to a plain LLM-only correction step. The magnitude of the WER gap (0.881 vs 0.820) is modest in absolute terms but statistically robust, and is accompanied by consistent drops in BLEU, ROUGE-L, and BERTScore. Taken together with the ten-consultation pilot, the full 57-consultation run shows that, under the current ASR quality and retrieval setup, clinical dense RAG does not recover phonetic errors from text context and instead tends to add clinically plausible but locally incorrect material.

Table 7: Mean WER on PriMock57 (n=10). ✓ = best condition.

Condition	WER ↓	Notes
C1 (Whisper)	0.872	ASR baseline
C2b (Mistral) ✓	**0.869**	LLM-only; best overall
C2a (LLaMA)	0.925	
BM25 Lexical		
C3-Lex-Gen	0.960	
C3-Lex-Rel	1.033	
C3-Lex-Mat	0.997	
Dense Embedding		
C4-Den-Gen	0.902	Best RAG condition
C4-Den-Mat	0.991	Leave-one-out PriMock57 (domain-matched)

Table 8: Mean metrics, PriMock57 full (n=57).

Condition	WER ↓	BLEU ↑	RG-L ↑	BERT ↑
C1 (Whisper)	0.821	**0.495**	**0.498**	0.838
C2b (Mistral)	**0.820**	0.036	0.296	**0.840**
C4-Den-Mat	0.881	0.024	0.201	0.826

Table 9: Wilcoxon results, C4-Den-Mat vs C2b, PriMock57 full (n=57).

Metric	p-value	Sig.
WER	<0.001	Yes
BLEU	0.004	Yes
ROUGE-L	<0.001	Yes
BERTScore	<0.001	Yes

5.7 Hypothesis Summary

- **H1 (RAG vs. LLM-only):** supported on TED; C4-Den-Gen and C4-Den-Mat beat C2b, while C3-Lex-Gen is worse (retrieval quality matters).
- **H2 (domain-specific vs. generic):** supported on TED; C4-Den-Mat is best (WER 0.140, ROUGE-L 0.905).
- **H3 (dense vs. lexical):** supported on TED (62% WER reduction, $p < 0.01$); on MTS, WER is not significant ($p = 0.071$) but BLEU/ROUGE-L/ BERTScore are ($p < 0.05$).
- **H4 (RAG on noisy clinical):** not supported on PriMock57. C4-Den-Mat is significantly worse than C2b on all four metrics (WER, BLEU, ROUGE-L, BERTScore; $p < 0.01$) and degrades DISEASE F1; generic retrieval is safer for biomedical entities.

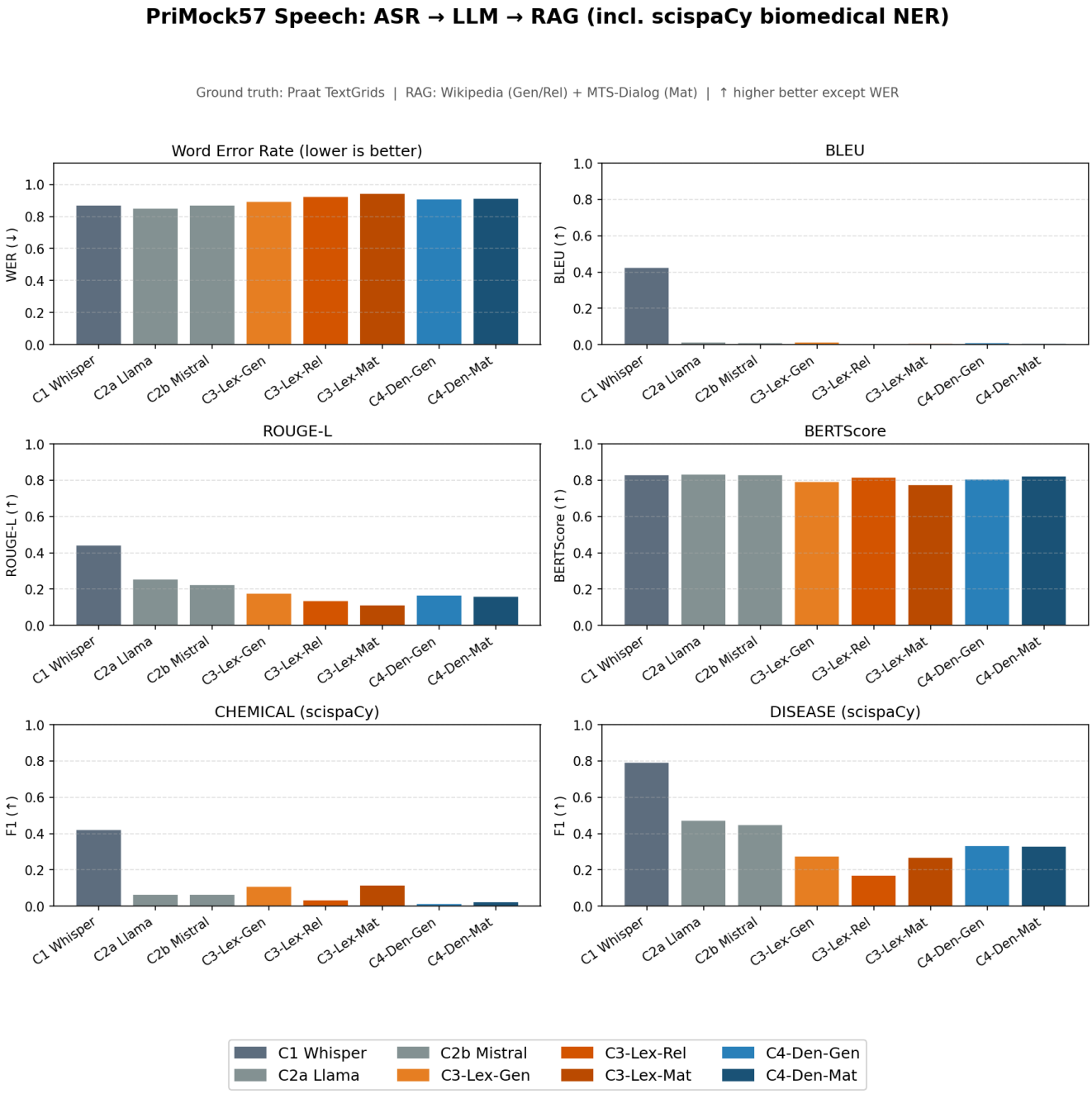


Figure 6: PriMock57: WER, BLEU, ROUGE-L, BERTScore, CHEMICAL F1, DISEASE F1 over seven corrected conditions (n=10). C1 leads on BLEU, ROUGE-L, NER; C2b is numerically best on WER and BERTScore (C2b and C1 are not statistically distinguished on WER); C4-Den-Mat is worse than C2b on all four text metrics. (Condition labels as in Table 1.)

6 Discussion

6.1 Retrieval Corpus and Modality Effects

Results show a strong interaction between retrieval modality and corpus scope. On TED-LIUM, generic lexical retrieval (C3-Lex-Gen) substantially worsens performance (WER 0.516), whereas dense retrieval on the same corpus (C4-Den-Gen) reduces WER by 62% ($p < 0.01$). Within BM25, corpus matters (C3-Lex-Rel 0.362 vs C3-Lex-Gen; BERTScore $p = 0.70$). Domain alignment amplifies gains: C4-Den-Mat 54% WER reduction over C2b, C4-Den-Gen 36%. PriMock57 C4-Den-Mat uses leave-one-out PriMock57 (domain-matched), same corpus as C3-Lex-Mat.

These patterns reflect different underlying acoustic regimes. On TED-LIUM, Whisper-tiny already achieves WER 0.079 under near-ideal conditions (clean, single-speaker, rehearsed speech). Any post-correction step that changes tokens risks mismatching the reference and inflating WER even when meaning is preserved. Dense RAG can afford mild paraphrase here, as shown by BERTScore remaining above 0.93 across corrected conditions, whereas BM25 lexical retrieval tends to inject irrelevant surface forms.

PriMock57 shows the opposite: baseline WER 0.872 leaves ample headroom, but neither lexical nor dense RAG clearly improves on Mistral-only. The 14% relative WER degradation of C4-Den-Mat vs C2b indicates that consultation- non-specific clinical passages can actively hurt performance when phonetic errors dominate and the model cannot rely on acoustic cues.

The PriMock57 degradation is unlikely to be an artefact. Four considerations support this. First, the error type differs structurally from MTS-Dialog: MTS noise is randomly injected at the word level, leaving contextual signals intact so a language model can often reconstruct the intended word from surrounding context. PriMock57 errors are phonetically grounded — Whisper has already transcribed a wrong word that is acoustically plausible, and the model receives only that wrong string with no acoustic signal to indicate it is incorrect. Text-context reconstruction is therefore fundamentally less tractable on phonetic errors than on random noise. Second, C3-Lex-Mat (WER 0.997) corroborates the finding from a different retrieval path: leave-one-out PriMock57 notes constitute the most tightly matched retrieval corpus available, yet performance is still worse than LLM-only — if corpus mismatch alone were responsible, this condition should outperform C4-Den-Mat, but it does not. Third, the mechanism is coherent: retrieved clinical passages from other consultations introduce plausible-but-locally-incorrect drug names and diagnoses, which the model substitutes into the transcript. Fourth, the pipeline and evaluation are standard and no obvious implementation error would produce this pattern consistently across all four metrics.

6.2 Entity Preservation in Clinical Dialogues

BERTScore and WER diverge in clinical conditions; NER explains why. The clinical-corpus penalty arises because dense retrieval returns semantically related clinical passages containing plausible medical entities; when the LLM encounters uncertain ASR tokens it often substitutes those entities into the transcript, preserving embedding-based similarity while degrading entity-level fidelity. General spaCy drops with Mistral (0.650–0.681 vs 0.766); LLaMA-Only (0.740) highest. ScispaCy: LLM and generic dense (C4-Den-Gen) preserve or improve DISEASE F1; with clinical corpora, C4-Den-Mat (DIS 0.837) is worse than C4-Den-Gen ($p = 0.0098$). Plausible-but-wrong entity substitutions from retrieved passages are invisible to BERTScore (0.93–0.96) but visible in DISEASE F1 (0.744–0.937); Table 5 breaks down by type.

The clinical corpus penalty arises from the interaction between dense retrieval and generative reformulation. Dense retrievers return semantically related clinical passages containing medically plausible entities. When the language model encounters uncertain tokens in the ASR transcript, it often replaces them with entity strings from the retrieved context. These substitutions maintain high semantic similarity but can introduce clinically incorrect entities. Because embedding-based metrics such as BERTScore reward semantic similarity, they remain largely invisible to standard evaluation metrics but are detected by biomedical NER.

Some C4-Den-Mat outputs had multiple `TRANSCRIPT` blocks (verbosity; WER > 1.0), likely prompt-following failure.

The PriMock57 NER behaviour illustrates a subtle evaluation issue. Raw Whisper often preserves surface forms from the reference transcription, so C1 can lead on BLEU, ROUGE-L, and NER F1 even when its clinical interpretations are no better than corrected outputs. LLM and RAG conditions frequently substitute formal terminology for lay phrasing ("hypertension" for "high blood pressure", "diabetes mellitus" for "diabetes"); exact-match NER penalises these substitutions even when they are semantically appropriate. For that reason, NER F1 on PriMock57 is best read as a measure of lexical fidelity to the reference note rather than a direct proxy for clinical correctness.

6.3 Error Analysis

Ten MTS-Dialog dialogues were manually inspected to characterise frequent failure modes. Table 10a lists

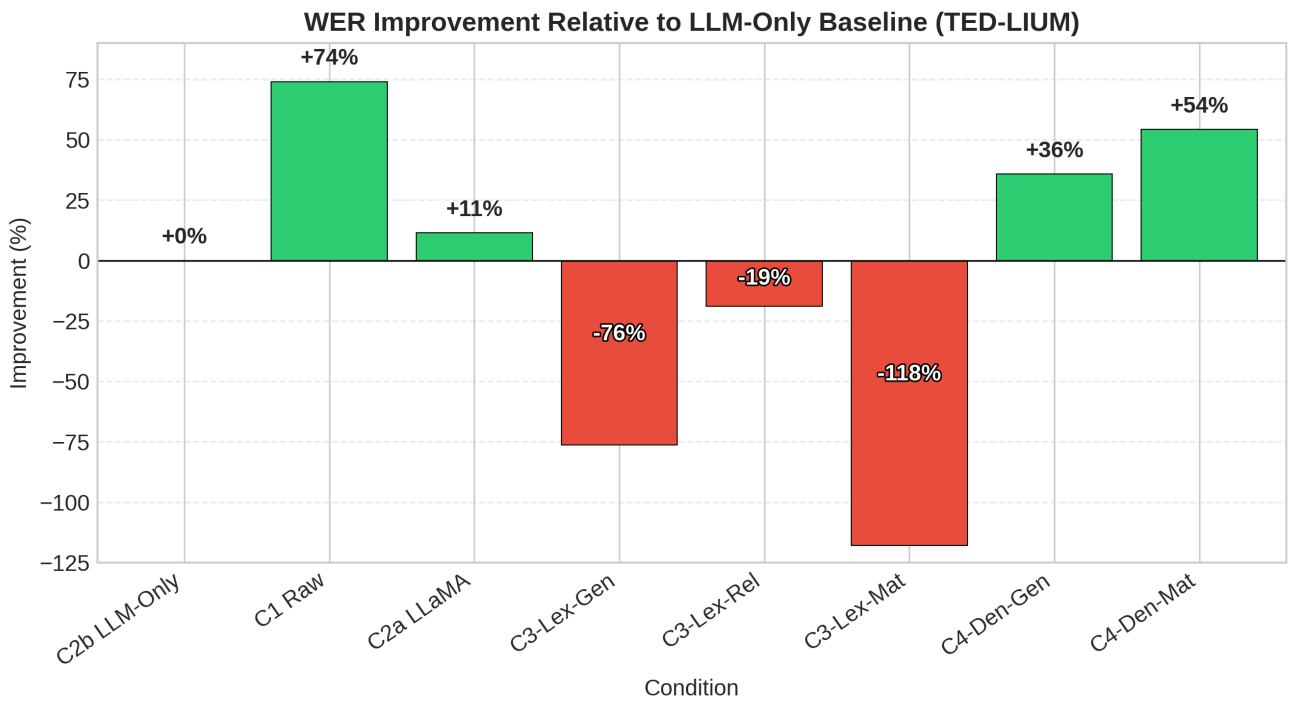


Figure 7: WER change relative to LLM-Only (TED-LIUM). Positive = WER reduction; negative = degradation.

examples where domain-matched dense retrieval helped by restoring omitted tokens or correcting deletions. Table 10b lists typical clinical-corpus penalty cases: entity swaps (one diagnosis replaced with a different plausible diagnosis), added severity (for example, "diabetes" → "uncontrolled diabetes mellitus"), and hallucinated history. These cases show that retrieved passages may supply medically plausible lexical items which the LLM substitutes for uncertain ASR tokens – improving surface fluency but damaging entity fidelity. The hallucination case (Dialogue 434) is the most clinically serious failure. The patient explicitly stated no family history of major diseases, yet C4-Den-Mat inserted several sentences of fabricated neurological history, plausibly templated from a retrieved clinical note. This is not a paraphrase error — it is factually incorrect clinical documentation introduced by retrieval. The entity-swap case (sarcoidosis for autoimmune disease) is subtler but equally problematic: both are disease entities so BERTScore registers little difference, but the two conditions are clinically distinct. The register-shift case (hypertension for high blood pressure) illustrates the opposite limitation: the substitution is clinically acceptable but penalised by exact-match NER because the entity string differs. No single metric captures clinical fidelity fully, which motivates entity-aware evaluation as a minimum requirement for clinical ASR assessment.

7 Limitations and Future Work

ASR model size. All experiments use Whisper-tiny, which produces a WER of 0.872 on PriMock57. Larger ASR models such as Whisper-medium or Whisper-large-v3 would likely produce cleaner transcripts and may reduce the difficulty of the correction task.

Simulated vs. real noise. Results on PriMock57 indicate that real clinical speech contains phonetically motivated ASR errors that are difficult to correct using text-only methods. Retrieval-based reformulation cannot fully recover the intended word when the underlying acoustic signal is severely misrecognised.

Evaluation. The error analysis in §6.3 is based on manual inspection and does not include inter-annotator agreement. Assessing biomedical entity correctness may require specialist annotation. In addition, the TED-LIUM analysis ($n = 10$) is exploratory. All experimental conditions were evaluated using the same language model (Mistral-7B).

Despite these limitations, the main result remains consistent: domain-matched retrieval reduces biomedical entity fidelity while semantic similarity metrics remain largely unchanged.

Future work. Several directions could address this failure mode. First, entity-aware retrieval, which restricts

(a) C4-Den-Mat beats C2b.

Error	C2b	C4-Den-Mat
Abbrev. drop	. . . *was treated*.	. . . *treated with ECT*.
Numeric drop	I have *children*.	I have *five children*.
Age error	You're *one year old*?	You're *61*?
Missing word	I was *and raised* in. . .	I was *born and raised*. . .

(b) C4-Den-Mat adds errors that C2b does not.

Error	Ground truth	C4-Den-Mat
Severity added	I have *diabetes*.	I have *uncontrolled diabetes mellitus*.
Entity swap	*autoimmune disease*	*sarcoidosis + depression*
Hallucination	No family history.	[Full paragraph of fabricated epilepsy history.]
Register shift	*high blood pressure*	*hypertension*

Table 10: Error analysis: (a) RAG helps; (b) clinical corpus penalty.

retrieved passages to those sharing entities with the ASR hypothesis, could reduce the observed drop in DISEASE F1. Second, prompt-level safeguards (e.g. instructions discouraging entity substitution) may help prevent the introduction of incorrect medical terms during reformulation.

Additional directions include human evaluation of transcript usefulness, experiments with larger ASR models (Whisper-medium or Whisper-large-v3), full biomedical NER evaluation on PriMock57, and investigation of low-resource speech settings [7]. One practical application is a voice-note correction pipeline in which a user's message history serves as the retrieval corpus. The clinical corpus penalty observed in §6.2 suggests that retrieval in such systems is best restricted to entity-matched or consultation-specific documents.

Graph-based retrieval approaches such as GraphRAG [17] and recent medical GraphRAG systems [18–21] may further reduce entity substitutions by introducing stronger structural constraints during retrieval.

Data and code availability. TED-LIUM and MTS-Dialog are publicly available datasets. PriMock57 can be obtained from the authors' repository (BabylonHealth LFS). Evaluation scripts and experimental configurations are available from the author on request.

Implementation environment. Experiments were run locally on a Fedora Linux workstation with a single NVIDIA GeForce RTX 5060 Laptop GPU (8 GB VRAM) and CPU-only evaluation scripts. The full ASR–RAG project, including experiment scripts, configuration files, and analysis notebooks, is released alongside this paper to support exact reproducibility.

8 Conclusion

This study shows that corpus scope mediates the effectiveness of retrieval-augmented correction. On clean, same-domain speech (TED-LIUM), narrow domain-matched retrieval yields substantial improvements. On noisy clinical data, however, domain-relevant or domain-matched retrieval can perform worse than generic retrieval or even LLM-only reformulation.

In particular, domain-focused retrieval introduces plausible but incorrect biomedical entity substitutions, which significantly reduce DISEASE F1 ($p < 0.01$) while leaving semantic similarity metrics such as BERTScore largely unchanged. As a result, standard semantic metrics may fail to capture clinically important errors.

This pattern holds for both MTS-Dialog (n=100) and PriMock57 (n=57). On TED-LIUM, domain-matched dense RAG reduces WER by 54% relative to LLM-only correction, while generic lexical RAG increases WER by 69%. Dense retrieval outperforms BM25 by 62% ($p < 0.01$). These improvements, however, do not transfer to PriMock57, where phonetic ASR errors remain resistant to text-only correction.

Overall, RAG-based transcript reformulation is beneficial when retrieval quality and corpus alignment are high, but it can be harmful when retrieved passages are only topically related rather than consultation-specific.

For practitioners, the main implication is caution when using clinical corpora. Domain-relevant retrieval is not automatically beneficial; without strong constraints it can introduce entity substitutions that surface-level metrics fail to detect. In practice, RAG-based ASR correction is applied only when the retrieval corpus is tightly linked to the consultation context (for example, the patient's own medical record or note history). Even in such cases, generic dense retrieval or LLM-only reformulation may remain safer defaults when entity fidelity is critical.

Entity-aware retrieval and graph-based constraints repres-

ent promising directions for future work. However, until these approaches are validated, they should currently be treated as experimental rather than reliable plug-and-play solutions.

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